

HW 1.2a

Name(s): _____

1. If Saturn lies 9.6 A.U. from the Sun, what is the distance between the Sun and Saturn in meters?

2. If Earth lies 1 A.U. from the Sun, and light travels at a speed of 300,000 kilometers per second, how many minutes does it take for sunlight emitted by the Sun to reach Earth?

3. Northwest Academy lies 7.25 kilometers from Mount Tabor. How many meters is that distance? How many centimeters?

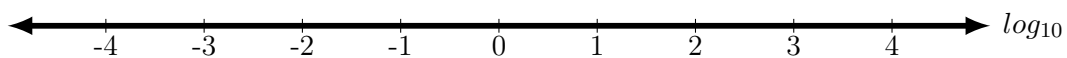
4. Write the following numbers in scientific notation
 - A. 12,300,000
 - B. 0.000487
 - C. 58,600
 - D. - 0.00000972

5. Convert the following numbers from scientific notation back into normal notation
 - A. 3.2×10^5
 - B. -6.4×10^{-3}
 - C. $1.18e5$
 - D. 4.9×10^6

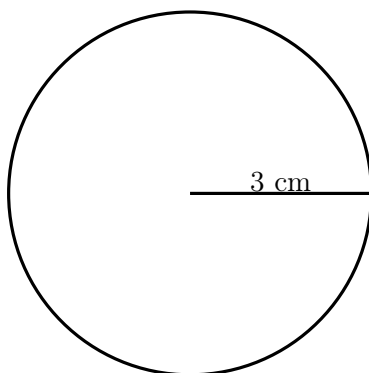
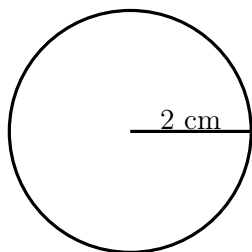
6. The following sizes are given as decimals, write them using the SI prefixes
 - A. Width of DNA, 0.0000000021 meters
 - B. Distance to the moon, 384,400,000 meters
 - C. Diameter of Jupiter, 139,820,000 meters
 - D. Distance to Pluto, 5,984,000,000,000 meters

7. Find the order of magnitude of the following sizes and plot them on this scale

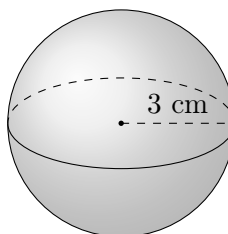
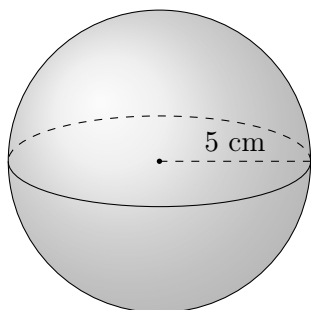
Object	Size
Human Hair	0.0002 meters
Football Field	98 meters
Cat	0.4 meters
Empire State Building	443 meters
Width of Oregon	400,000 meters



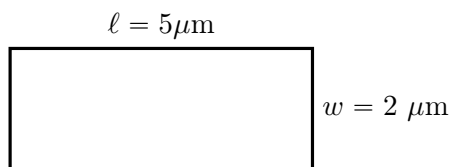
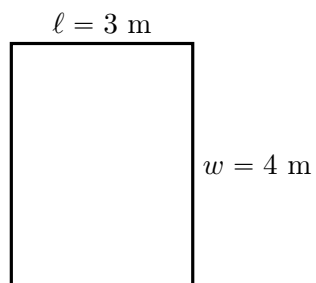
8. For each circle find the Circumference and Area.



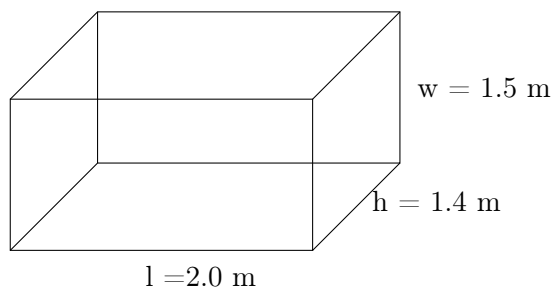
9. For each sphere, find the Surface area and Volume.



10. For each rectangle, find the perimeter and area



11. For the rectangular prism, find the surface area and volume



12. What are the seven base units of the *SI* system of measurement?

13. How are the units for meters, kilograms, and seconds defined?